EXPERIMENT 9

AIM

To determine Young's modulus of the material of a given wire by using Searle's apparatus.

APPARATUS AND MATERIAL REQUIRED

Searle's apparatus, slotted weights, experimental wire, screw gauge and spirit level.

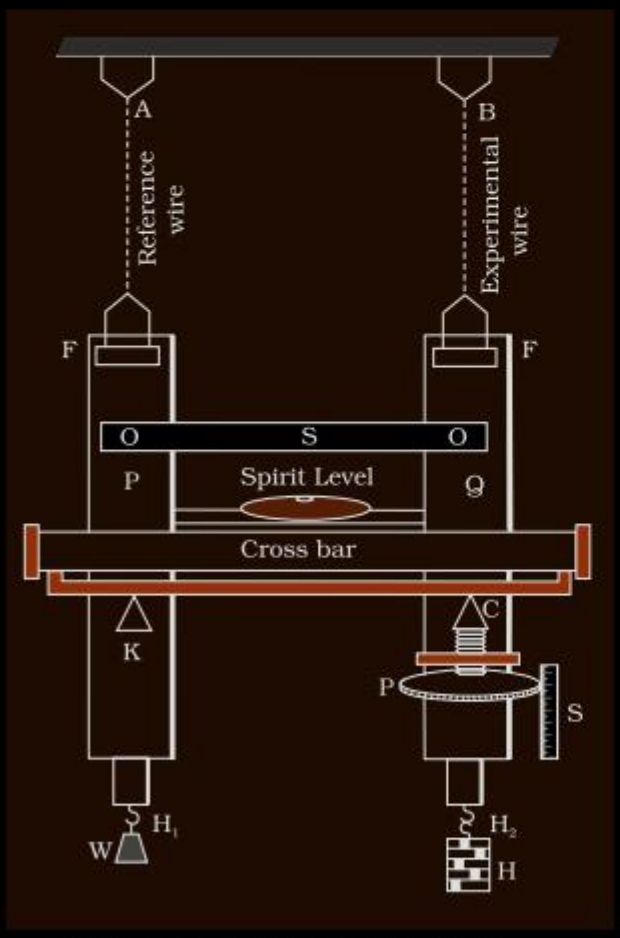


Fig. E 9.1: *Searle's apparatus for determination of Y*

SEARLE'S APPARATUS

It consists of two metal frames P and Q hinged together such that they can move relative to each other in vertical direction (Fig. E9.1).

A spirit level is supported on a rigid crossbar frame which rests on the tip of a micrometer screw C at one end and a fixed knife edge K at the other. Screw C can be moved vertically. The micrometer screw has a disc having 100 equal divisions along its circumference. On the side of it is a linear scale S, attached vertically. If there is any relative displacement between the two frames, P and Q, the spirit level no longer remains horizontal and the bubble of the spirit level is displaced from its centre. The crossbar can again be set horizontal with the help of micrometer screw and the spirit level. The distance through which the screw has to be moved gives the relative displacement between the two frames.

The frames are suspended by two identical long wires of the same material, from the same rigid horizontal support. Wire B is called the experimental wire and wire A acts as a reference wire. The frames, P and Q, are provided with hooks H_1 and H_2 at their lower ends from which weights are suspended. The hook H_1 attached to the reference wire carries a constant weight *W* to keep the wire taut.

To the hook H_2 is attached a hanger on which slotted weights can be placed to apply force on the experimental wire.

PRINCIPLE

The apparatus works on the principle of Hookes' Law. If l is the extension in a wire of length L and radius r due to force F (=Mg), the Young's modulus of the material of the given wire, Y, is

$$Y = \frac{MgL}{\pi r^2 l}$$

PROCEDURE

1. Suspend weights from both the hooks so that the two wires are stretched and become free from any kinks. Attach only the constant weight W on the reference wire to keep it taut.
2. Measure the length of the experimental wire from the point of its support to the point where it is attached to the frame.
3. Find the least count of the screw gauge. Determine the diameter of the experimental wire at about 5 places and at each place in two mutually perpendicular directions. Find the mean diameter and hence the radius of the wire.
4. Find the pitch and the least count of the miocrometer screw attached to the frame. Adjust it such that the bubble in the spirit level is exactly in the centre. Take the reading of the micrometer.
5. Place a load on the hanger attached to the experimental wire and increase it in steps of 0.5 kg. For each load, bring the bubble of the spirit level to the centre by adjusting the micrometer screw and then note its reading. Take precautions to avoid backlash error.
6. Take about 8 observations for increasing load.
7. Decrease the load in steps of 0.5 kg and each time take reading on micrometer screw as in step 5.

OBSERVATIONS

Length of the wire (L) = ...

Pitch of the screw gauge = ...

No. of divisions on the circular scale of the screw gauge = ...

Least count (L.C.) of screw gauge = ...

Zero error of screw gauge = ...

Table E 9.1: Measurement of diameter of wire

S. No.	Reading along any direction			Reading along perpendicular direction			Mean diameter $d = \frac{d_1 + d_2}{2}$ (cm)
	Main scale reading S (cm)	Circular scale reading n	Diameter $d_1 = S + n$ L.C.	Main scale reading S (cm)	Circular scale reading n	Diameter $d_2 = S + n$ L.C. (cm)	
1							
2							
3							
4							
5							

Mean diameter (corrected for zero error) = ...

Mean radius = ...

MEASUREMENT OF EXTENSION *l*

Pitch of the micrometer screw = ...

No. of divisions on the circular scale = ...

Least count (L.C.) of the micrometer screw = ...

Acceleration due to gravity, g = ...

Table E 9.2: Measurement of extension with load

S. No.	Load on experimental wire M	Micrometer reading		Mean reading $\left(\frac{x + y}{2}\right)$ (cm)
	(kg)	Load increasing x (cm)	Load decreasing y (cm)	
1	0.5			a
2	1.0			b
3	1.5			c
4	2.0			d
5	2.5			e
6	3.0			f
7	3.5			g
8	4.0			h

CALCULATION

Observations recorded in Table E 9.2 can be utilised to find extension of experimental wire for a given load, as shown in Table E 9.3.

Table E 9.3: Calculating extension for a given load

S. No.	Mean extension (cm)	Load (kg)	Mean extension	Extension l' for 1.5 kg
0.5		2.0		d – a
1.0		2.5		e – b
1.5		3.0		f – c

$$\therefore \text{Mean } l = \frac{(a-d)+(b-c)+(c-f)}{3}$$

$$= \ldots \text{ cm for } 1.5 \text{ kg}$$

Young's modulus, Y, of experimental wire $Y = \frac{MgL}{\pi r^2 l} = \ldots \text{N/m}^2$

GRAPH

The value of Y can also be found by plotting a graph between l and Mg. Draw a graph with load on the x-axis and extension on the y-axis. It should be a straight line. Find the slope $= \frac{\Delta l}{\Delta M}$ of the line. Using this value, find the value of Y.

RESULT

The Young's modulus Y of the material of the wire

(using half table method) $= Y \pm \Delta Y \text{ N/m}^2$

(using graph) $= Y \pm \Delta Y \text{ N/m}^2$

ERROR

Uncertainty, ΔM, in the measurement of M can be determined by a beam/physical balance using standard weight box/or by using water bottles of fixed capacity.

Find the variation in M for each slotted weight of equal mass say ΔM_1 and ΔM_2. Find the mean of these ΔM. This is the uncertainity (ΔM) in M.

ΔL – the least count of the scale used for measuring L.

Δr – the least count of the micrometer screw gauge used for measuring r.

Δl – least count of the device used for measuring extension.

PRECAUTION

1. Measure the diameter of the wire at different positions, check for its uniformity.
2. Adjust the spirit level only after sufficient time gap following each loading/unloading.

SOURCES OF ERROR

1. The diameter of the wire may alter while loading.
2. Backlash error of the device used for measuring extension.
3. The nonuniformity in thickness of the wire.

DISCUSSION

Which of the quantities measured in the experiment is likely to have maximum affect on the accuracy in measurement of Y (Young's modulus).

SELF ASSESSMENT

1. If the length of the wire used is reduced what will be its effect on
   (a) extension on the wire and (b) stress on the wire.
2. Use wire of different radii (r_1, r_2, r_3) but of same material in the above experimental set up. Is there any change in the value of Young's modulus of elasticity of the material? Discuss your result.

SUGGESTED ADDITIONAL EXPERIMENTS/ACTIVITIES

1. Repeat the experiment with wires of different materials, if available.
2. Change the length of the experimental wire, of same material and study its effect on the Young's modulus of elasticity of the material.

EXPERIMENT 17

AIM

To determine the specific heat capacity of a given (i) and solid (ii) a liquid by the method of mixtures.

APPARATUS AND MATERIAL REQUIRED

Copper calorimeter with lid, stirrer and insulating cover (the lid should have provision to insert thermometer in addition to the stirrer), two thermometers (0 C to 100 C or 110 C with a least count of 0.5 C), a solid, preferably metallic (brass/copper/steel/aluminium) cylinder which is insoluble in given liquid and water, given liquid, two beakers (100 mL and 250 mL), a heating device (heater/hot plate/gas burner); physical balance, spring balance with weight box (including fractional weights), a piece of strong non-flexible thread (25-30 cm long), water, laboratory stand, tripod stand and wire gauze.

PRINCIPLE / THEORY

For a body of mass m and specific heat s, the amount of heat Q lost/gained by it when its temperature falls/rises by Δt is given by

$$\Delta Q = ms\,\Delta t \qquad \text{(E 17.1)}$$

Specific heat capacity: It is the amount of heat required to raise the temperature of unit mass of a substance through 1 C. Its S.I unit is $Jkg^{-1}\,K^{-1}$.

Principle of Calorimetry: If bodies of different temperatures are brought in thermal contact, the amount of heat lost by the body at higher temperature is equal to the amount of heat gained by the body at lower temperature, at thermal equilibrium, provided no heat is lost to the surrounding.

(a) Specific heat capacity of given solid by method of mixtures

PROCEDURE

1. Set the physical balance and make sure there is no zero error.
2. Weigh the empty calorimeter with stirrer and lid with the physical balance/spring balance. Ensure that calorimeter is clean and dry.

Note the mass m_1 of the calorimeter. Pour the given water in the calorimeter. Make sure that the quantity of water taken would be sufficient to completely submerge the given solid in it. Weigh the calorimeter with water along with the stirrer and the lid and note its mass m_2. Place the calorimeter in its insulating cover.

3. Dip the solid in water and take it out. Now shake it to remove water sticking to its surface. Weigh the wet solid with the physical balance and note down its mass m_3.

4. Tie the solid tightly with the thread at its middle. Make sure that it can be lifted by holding the thread without slipping.

   Place a 250 mL beaker on the wire gauze kept on a tripod stand as shown in the Fig. E 17.1(a). Fill the beaker up to the half with water. Now suspend the solid in the beaker containing water by tying the other end of the thread to a laboratory stand. The solid should be completely submerged in water and should be atleast 0.5 cm below the surface. Now heat the water with the solid suspended in it [Fig. E 17.1 (a)].

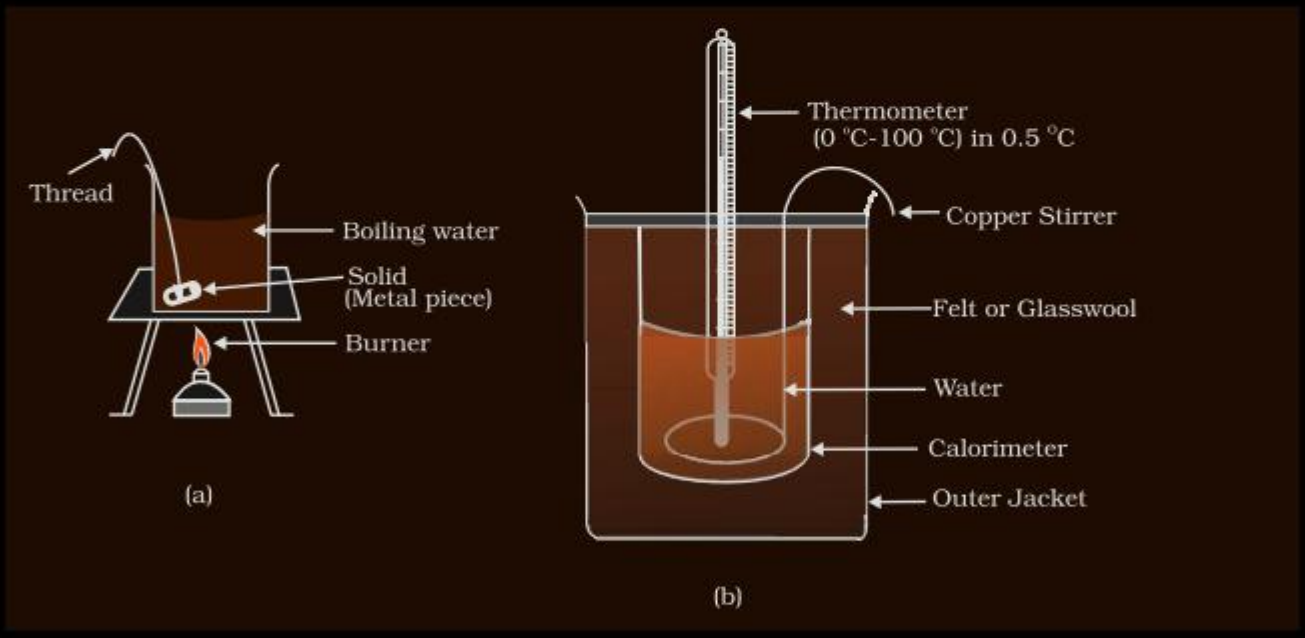


Fig. E 17.1: *Experimental setup for determining specific heat of a given solid*

5. Note the least count of the thermometer. Measure the temperature of the water taken in the calorimeter. Record the temperature t_1 of the water.

6. Let the water in the beaker boil for about 5-10 minutes. Now measure the temperature t_2 of the water with the other thermometer and record the same. Holding the solid with the thread tied to it,

remove it from the boiling water, shake it to remove water sticking on it and quickly put it in the water in the calorimeter and replace the lid immediately (Fig. E 17.1 (b)). Stir the water with the stirrer. Measure the temperature of the water once equilibrium is attained, that is, temperature of the mixture becomes constant. Record this temperature as t_3.

OBSERVATIONS

Mass of the empty calorimeter with stirrer (m_1) = ... g

Mass of the calorimeter with water (m_2) = ... g

Mass of solid (m_3) = ... g

Initial temperature of the water (t_1) = ... C = ... K

Temperature of the solid in boiling water (t_2) = ... C = ... K

Temperature of the mixture (t_3) = ... C

Specific heat capacity of material of calorimeter s_1 = ... Jkg^{-1} C^{-1} ($Jkg^{-1}K^{-1}$)

Specific heat capacity of water (s) = ... $Jkg^{-1}K^{-1}$

CALCULATIONS

1. Mass of the water in calorimeter ($m_2 - m_1$) = ... g = ... kg
2. Change in temperature of liquid and calorimeter ($t_3 - t_1$) = ... C
3. Change in temperature of solid ($t_2 - t_3$) == ... C

   Heat given by solid in cooling from t_2 to t_3.

   = Heat gained by liquid in raising its temperature from t_1 to t_3 + heat gained by calorimeter in raising its temperature from t_1 to t_3.

   $m_3 s_o (t_2 - t_3) = (m_2 - m_1)\, s\, (t_2 - t_1) + m_1 s_1 (t_3 - t_1)$

   $$s_o = \frac{(m_2 - m_1)s(t_2 - t_1) + m_1 s_1 (t_3 - t_1)}{m_3 (t_2 - t_3)} = \text{... J kg}^{-1}\ \text{C}^{-1}$$

(b) Specific heat capacity of given liquid by method of mixtures

PROCEDURE

1. Set the phyiscal balance and make sure there is no zero error.
2. Weigh the empty calorimeter with stirrer and lid with the physical balance/spring balance. Ensure that calorimeter is clean and dry. Note the mass m_1 of the calorimeter. Pour the

given liquid in the calorimeter. Make sure that the quantity of liquid taken would be sufficient to completely submerge the solid in it. Weigh the calorimeter with liquid along with the stirrer and the lid and note its mass m_2. Place the calorimeter in its insulating cover.

3. Take a metallic cylinder whose specific heat capacity is known. Dip it in water in a container and shake it to remove the water sticking to its surface. Weigh the wet solid with the physical balance and note down its mass m_3.

4. Tie the solid tightly with the thread at its middle. Make sure that it can be lifted by holding the thread without slipping.

   Place a 250 mL beaker on the wire gauze kept on a tripod stand as shown in Fig. E 17.1(a). Fill the beaker up to half with water. Now suspend the solid in the beaker containing water by tying the other end of the thread to a laboratory stand. The solid should be completely submerged in water and should be atleast 0.5 cm below the surface. Now heat the water with the solid suspended in it [Fig. E 17.1(a)].

5. Note the least count of the thermometer. Measure the temperature of the water taken in the calorimeter. Record the temperature t_1 of the water.

6. Let the liquid in the beaker boil for about 5-10 minutes. Now measure the temperature t_2 of the liquid with the other thermometer and record the same. Holding the solid with the thread tied to it remove it from the boiling water, shake it to remove water sticking on it and quickly put it in the liquid in the calorimeter and replace the lid immediately [Fig. E 17.1(b)]. Stir it with the stirrer. Measure the temperature of the liquid once equilibrium is attained, that is, temperature of the mixture becomes constant. Record this temperature as t_3.

OBSERVATIONS

Mass of the empty calorimeter with stirrer (m_1) = ... g

Mass of the calorimeter with liquid (m_2) = ... g

Mass of solid (m_3) = ... g

Initial temperature of the liquid (t_1) = C = ... K

Temperature of the solid in boiling water (t_2) = C = ... K

Temperature of the mixture (t_3) = C = ... K

Specific heat capacity of material of calorimeter s_1 = ... Jkg^{-1} C^{-1} (Jkg^{-1} K^{-1})

Specific heat capacity of solid (s_0) = ... Jkg^{-1} K^{-1}

CALCULATIONS

1. Mass of the liquid in calorimeter $(m_2 - m_1)$ = ... g = ... kg
2. Change in temperature of liquid and calorimeter $(t_3 - t_1)$ = ... C
3. Change in temperature of solid $(t_2 - t_3)$ = ... C

   Heat given by solid in cooling from t_2 to t_3.

   = Heat gained by liquid in raising its temperature from t_1 to t_3 + heat gained by calorimeter in raising its temperature from t_1 to t_3.

   $m_3 s_o (t_2 - t_3) = (m_2 - m_1)\, s\, (t_2 - t_1) + m_1 s_1 (t_3 - t_1)$

   $$s = \frac{m_3 s_0 (t_2 - t_3) - m_1 s_1 (t_3 - t_1)}{(m_2 - m_1)(t_2 - t_1)} = \ldots \text{ J kg}^{-1}\ \text{C}^{-1}$$

RESULT

(a) The specific heat of the given solid is ... $Jkg^{-1} K^{-1}$ within experimental error.

(b) The specific heat of the given liquid is ... $Jkg^{-1} K^{-1}$ within experimental error.

PRECAUTIONS

1. Physical balance should be in proper working condition and ensure that there is no zero error.
2. The two thermometers used should be of the same range and least count.
3. The solid used should not be chemically reactive with the liquid used or water.
4. The calorimeter should always be kept in its insulated cover and at a sufficient distance from the source of heat and should not be exposed to sunlight so that it absorbs no heat from the surrounding.
5. The solid should be transferred quickly so that its temperature is same as recorded when it is dropped in the liquid.
6. Liquid should not be allowed to splash while dropping the solid in it in the calorimeter. It is advised that the solid should be lowered gently into the liquid with the help of the thread tied to it.
7. While measuring the temperature, the thermometers should always be held in vertical position. The line of sight should be perpendicular to the mercury level while recording the temperature.

SOURCES OF ERROR

1. Radiation losses cannot be completely eliminated.
2. Heat loss that takes place during the short period while transferring hot solid into calorimeter, cannot be accounted for.
3. Though mercury in the thermometer bulb has low specific heat, it absorbs some heat.
4. There may be some error in measurement of mass and temperature.

DISCUSSION

1. There may be some heat loss while transferring the solid, from boiling water to the liquid kept in the calorimeter. Heat loss may also occur due to time lapsed between putting of hot solid in calorimeter and replacing its lid.
2. The insulating cover of the calorimeter may not be a perfect insulator.
3. Error in measurement of mass of calorimeter, calorimeter with liquid and that of the solid may affect the calculation of specific heat capacity of the liquid.
4. Calculation of specific heat capacity of the liquid may also be affected by the error in measurement of temperatures.
5. Even though the metal piece is kept in boiling water, it may not have exactly the same temperature as that of boiling water.

SELF ASSESSMENT

1. What is water equivalent?
2. Why do we generally use a calorimeter made of copper?
3. Why is it important to stir the contents before taking the temperature of the mixture?
4. Is specific heat a constant quantity?
5. What is thermal equilibrium?

SUGGESTED ADDITIONAL EXPERIMENTS/ACTIVITIES

We can verify the principle of calorimetry, if specific heat capacity of the solid and the liquid are known.

EXPERIMENT 6

AIM

Using a Simple Pendulum plot L – T and L – T^2 graphs, hence find the effective length of second's pendulum using appropriate graph.

APPARATUS AND MATERIAL REQUIRED

Clamp stand; a split cork; a heavy metallic (brass/iron) spherical bob with a hook; a long, fine, strong cotton thread/string (about 2.0 m); stop-watch; metre scale, graph paper, pencil, eraser.

DESCRIPTION OF TIME MEASURING DEVICES IN A SCHOOL LABORATORY

The most common device used for measuring time in a school laboratory is a stop-watch or a stop-clock (analog). As the names suggest, these have the provision to start or stop their working as desired by the experimenter.

(a) Stop-Watch

Analog

A stop-watch is a special kind of watch. It has a multipurpose knob or button (B) for start/stop/back to zero position [Fig. E 6.1(b)]. It has two circular dials, the bigger one for a longer second's hand and the other smaller one for a shorter minute's hand. The second's dial has 30 equal divisions, each division representing 0.1 second. Before using a stop-watch you should find its least count. In one rotation, the seconds hand covers 30 seconds (marked by black colour) then in the second rotation another 30 seconds are covered (marked by red colour), therefore, the least count is 0.1 second.



Fig.E 6.1(a): *Stop - Watch*

(b) Stop-Clock

The least count of a stop-watch is generally about 0.1s [Fig. E 6.1(b)] while that of a stop-clock is 1s, so for more accurate measurement of time intervals in a school laboratory, a stop-watch is preferred. Digital stop-watches are also available now. These watches may be started by pressing the button and can be stopped by pressing the same button

once again. The lapsed time interval is directly displayed by the watch.



Fig.E 6.1(b): *Stop - Clock*

TERMS AND DEFINITIONS

1. **Second's pendulum:** It is a pendulum which takes precisely one second to move from one extreme position to other. Thus, its times period is precisely 2 seconds.
2. **Simple pendulum:** A point mass suspended by an inextensible, mass less string from a rigid point support. In practice a small heavy spherical bob of high density material of radius *r*, much smaller than the length of the suspension, is suspended by a light, flexible and strong string/thread supported at the other end firmly with a clamp stand. Fig. E 6.2 is a good approximation to an ideal simple pendulum.
3. **Effective length of the pendulum:** The distance L between the point of suspension and the centre of spherical bob (centre of gravity), $L = l + r + e$, is also called the effective length where l is the length of the string from the top of the bob to the hook, e, the length of the hook and r the radius of the bob.

PRINCIPLE

The simple pendulum executes **Simple Harmonic Motion** (SHM) as the acceleration of the pendulum bob is directly proportional to its displacement from the mean position and is always directed towards it.

The time period (T) of a simple pendulum for oscillations of small amplitude, is given by the relation

$$T = 2\pi\sqrt{L/g} \qquad \text{(E 6.1)}$$

where L is the length of the pendulum, and g is the acceleration due to gravity at the place of experiment.

Eq. (6.1) may be rewritten as

$$T^2 = \frac{4\pi^2 L}{g} \qquad \text{(E 6.2)}$$

PROCEDURE

1. Place the clamp stand on the table. Tie the hook, attached to the pendulum bob, to one end of the string of about 150 cm in length. Pass the other end of the string through two half-pieces of a split cork.

2. Clamp the split cork firmly in the clamp stand such that the line of separation of the two pieces of the split cork is at right angles to the line OA along which the pendulum oscillates [Fig. E 6.2(a)]. Mark, with a piece of chalk or ink, on the edge of the table a vertical line parallel to and just behind the vertical thread OA, the position of the bob at rest. Take care that the bob hangs vertically (about 2 cm above the floor) beyond the edge of the table so that it is free to oscillate.

3. Measure the effective length of simple pendulum as shown in Fig. E 6.2(b).

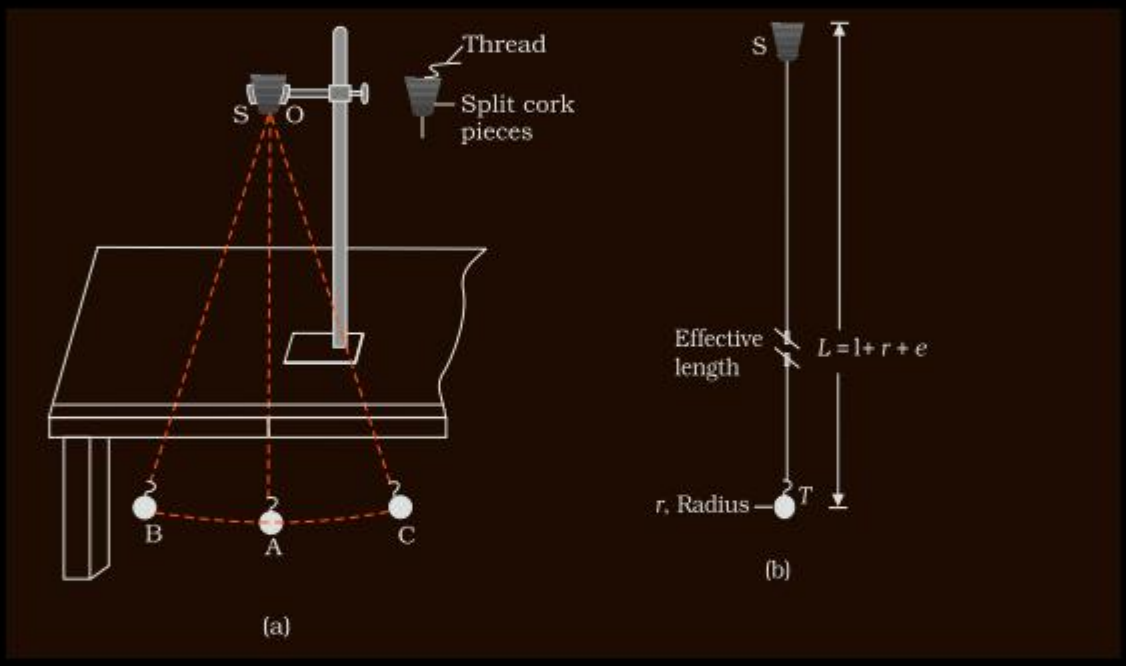


Fig.E 6.2 (a): *A simple pendulum; B and C show the extreme positions*

Fig.E 6.2 (b): *Effective length of a simple pendulum*

4. Displace the bob to one side, not more than 15 degrees angular displacement, from the vertical position OA and then release it gently. In case you find that the stand is shaky, put some heavy object on its base. Make sure that the bob starts oscillating in a vertical plane about its rest (or mean) position OA and **does not** (i) spin about its own axis, or (ii) move up and down while oscillating, or (iii) revolve in an elliptic path around its mean position.

5. Keep the pendulum oscillating for some time. After completion of a few oscillations, start the stop-watch/clock as the thread attached to the pendulum bob just crosses its mean position (say, from left to right). Count it as zero oscillation.

6. Keep on counting oscillations 1,2,3,…, n, everytime the bob crosses the mean position OA in the same direction (from left to right).

Stop the stop-watch/clock, at the count n (say, 20 or 25) of oscillations, i.e., just when n oscillations are complete. For better results, n should be chosen such that the time taken for n oscillations is 50 s or more. Read, the total time (t) taken by the bob for n oscillations. Repeat this observation a few times by noting the time for same number (n) of oscillations. Take the mean of these readings. Compute the time for one oscillation, i.e., the time period T (= t/n) of the pendulum.

7. Change the length of the pendulum, by about 10 cm. Repeat the step 6 again for finding the time (t) for about 20 oscillations or more for the new length and find the mean time period. Take 5 or 6 more observations for different lengths of penduLum and find mean time period in each case.

8. Record observations in the tabular form with proper units and significant figures.

9. Take effective length L along x-axis and T^2 (or T) along y-axis, using the observed values from Table E 6.1. Choose suitable scales on these axes to represent L and T^2 (or T). Plot a graph between L and T^2 (as shown in Fig. E 6.4) and also between L and T (as shown in Fig. E 6.3). What are the shapes of L–T^2 graph and L–T graph? Identify these shapes.

OBSERVATIONS

(i) Radius (r) of the pendulum bob (given) = ... cm

Length of the hook (given) (e) = ... cm

Least count of the metre scale = ... mm = ... cm

Least count of the stop-watch/clock = ... s

Table E 6.1: Measuring the time period T and effective length L of the simple pendulum

S. No.	Length of the string from the top of the bob to the point of suspension l	Effective length, L = (l+r+e)		Number of oscillations counted, n	Time for n oscillations t (s)				Time period T (= t/ n)
		(cm)	m		(i)...	(ii)	(iii)	Mean t (s)	s

PLOTTING GRAPH

(i) *L* vs *T* graphs

Plot a graph between L versus T from observations recorded in Table E 6.1, taking L along x-axis and T along y-axis. You will find that this graph is a curve, which is part of a parabola as shown in Fig. E 6.3.

(ii) *L* vs T^2 graph

Plot a graph between L versus T^2 from observations recorded in Table E 6.1, taking L along x-axis and T^2 along y-axis. You will find that the graph is a straight line passing through origin as shown in Fig. E. 6.4.

(iii) From the T^2 versus L graph locate the effective length of second's pendulum for $T^2 = 4s^2$.

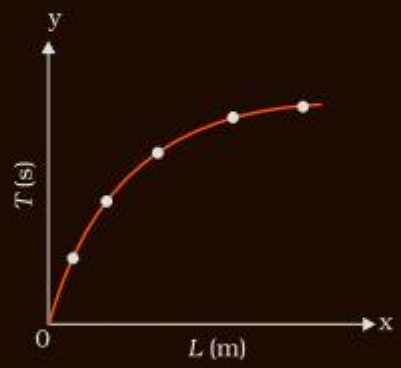


Fig. E 6.3: *Graph of L vs T*

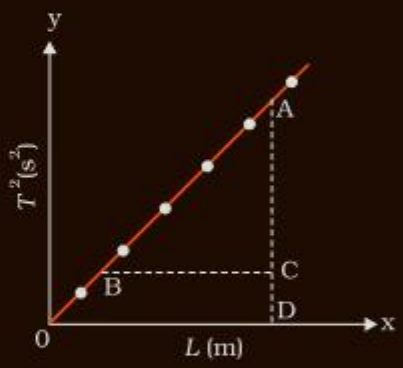


Fig. E 6.4: *Graph L vs T^2*

RESULT

1. The graph L versus T is curved, convex upwards.
2. The graph L versus T^2 is a straight line.
3. The effective length of second's pendulum from L versus T^2 graph is ... cm.

Note : The radius of bob may be found from its measured diameter with the help of callipers by placing the pendulum bob between the two jaws of (a) ordinary callipers, or (b) Vernier Callipers, as described in Experiment E 1.1 (a). It can also be found by placing the spherical bob between two parallel card boards and measuring the spacing (diameter) or distance between them with a metre scale.

DISCUSSION

1. The accuracy of the result for the length of second's pendulum depends mainly on the accuracy in measurement of effective length (using metre scale) and the time period T of the pendulum (using stop-watch). As the time period appears as T^2 in Eq. E 6.2, a small uncertainty in the measurement of T would result in appreciable error in T^2, thereby significantly affecting the result. A stop-watch with accuracy of 0.1s may be preferred over a less accurate stop-watch/clock.

2. Some personal error is always likely to be involved due to stop-watch not being started or stopped exactly at the instant the bob crosses the mean position. Take special care that you start and stop the stop-watch at the instant when pendulum bob just crosses the mean position in the same direction.

3. Sometimes air currents may not be completely eliminated. This may result in conical motion of the bob, instead of its motion in vertical plane. The spin or conical motion of the bob may cause a twist in the thread, thereby affecting the time period. Take special care that the bob, when it is taken to one side of the rest position, is released very gently.

4. To suspend the bob from the rigid support, use a thin, light, strong, unspun cotton thread instead of nylon string. Elasticity of the string is likely to cause some error in the effective length of the pendulum.

5. The simple pendulum swings to and fro in SHM about the mean, equilibrium position. Eq. (E 6.1) that expresses the relation between T and L as $T = 2\pi\sqrt{L/g}$, holds strictly true for small amplitude or swing θ of the pendulum.

   Remember that this relation is based on the assumption that sin $\theta \approx \theta$, (expressed in radian) holds only for small angular displacement θ.

6. Buoyancy of air and viscous drag due to air slightly increase the time period of the pendulum. The effect can be greatly reduced to a large extent by taking a small, heavy bob of high density material (such as iron/ steel/brass).

SELF ASSESSMENT

1. Interpret the graphs between L and T^2, and also between L and T that you have drawn for a simple pendulum.

2. Examine, using Table E 6.1, how the time period T changes as the

effective length L of a simple pendulum; becomes 2-fold, 4-fold, and so on.

3. How can you determine the value of 'g', acceleration due to gravity, from the T^2 vs L graph?

SUGGESTED ADDITIONAL EXPERIMENTS/ACTIVITIES

1. To determine 'g', the acceleration due to gravity, at a given place, from the $L - T^2$ graph, for a simple pendulum.

2. Studying the effect of size of the bob on the time period of the simple pendulum.

   [**Hint:** With the same experimental set-up, take a few spherical bobs of same material (density) but of different sizes (diameters). Keep the length of the pendulum the same for each case. Clamp the bobs one by one, and starting from a small angular displacement of about 10°, each time measure the time for 50 oscillations. Find out the time period of the pendulum using bobs of different sizes. Compensate for difference in diameter of the bob by adjusting the length of the thread.

   Does the time period depend on the size of the pendulum bob? If yes, see the order in which the change occurs.]

3. Studying the effect of material (density) of the bob on the time period of the simple pendulum.

   [**Hint:** With the same experimental set-up, take a few spherical bobs (balls) of different materials, but of same size. Keep the length of the pendulum the same for each case. Find out, in each case starting from a small angular displacement of about 10 , the time period of the pendulum using bobs of different materials,

   Does the time period depend on the material (density) of the pendulum bob? If yes, see the order in which the change occurs. If not, then do you see an additional reason to use the pendulum for time measurement.]

4. Studying the effect of mass of the bob on the time period of the simple pendulum.

   [**Hint:** With the same experimental set-up, take a few bobs of different materials (different masses) but of same size. Keep the length of the pendulum same for each case. Starting from a small angular displacement of about 10 find out, in each case, the time period of the pendulum, using bobs of different masses.

   Does the time period depend on the mass of the pendulum bob? If yes, then see the order in which the change occurs. If not, then do you see an additional reason to use the pendulum as a time measuring device.]

5. Studying the effect of amplitude of oscillation on the time period of the simple pendulum.

   [**Hint:** With the same experimental set-up, keep the mass of the bob and length of the pendulum fixed. For measuring the angular amplitude, make a large protractor on the cardboard and have a scale marked on an arc from 0 to 90 in units of 5 . Fix it on the edge of a table by two drawing pins such that its 0 - line coincides

with the suspension thread of the pendulum at rest. Start the pendulum oscillating with a very large angular amplitude (say 70) and find the time period T of the pendulum. Change the amplitude of oscillation of the bob in small steps of 5 or 10 and determine the time period in each case till the amplitude becomes small (say 5). Draw a graph between angular amplitude and T. How does the time period of the pendulum change with the amplitude of oscillation?

How much does the value of T for A = 10 differ from that for A= 50 from the graph you have drawn?

Find at what amplitude of oscillation , the time period begins to vary?

Determine the limit for the pendulum when it ceases to be a simple pendulum.]

6. Studying the effect on time period of a pendulum having a bob of varying mass (e.g. by filling the hollow bob with sand, sand being drained out in steps)

[**Hint:** The change in T, if any, in this experiment will be so small that it will not be possible to measure it due to the following reasons:

The centre of gravity (CG) of a hollow sphere is at the centre of the sphere. The length of this simple pendulum will be same as that of a solid sphere (same size) or that of the hollow sphere filled completely with sand (solid sphere).

Drain out some sand from the sphere. The situation is as shown in Fig. E. 6.5. The CG of bob now goes down to point say A. The effective length of the pendulum increases and therefore the T_A increases ($T_A > T_O$), some more sand is drained out, the CG goes down further to a point B. The effective length further increases, increasing T.

The process continues and L and T change in the same direction (increasing), until finally the entire sand is drained out. The bob is now a hollow sphere with CG shifting back to centre C. The time period will now become T_O again.]

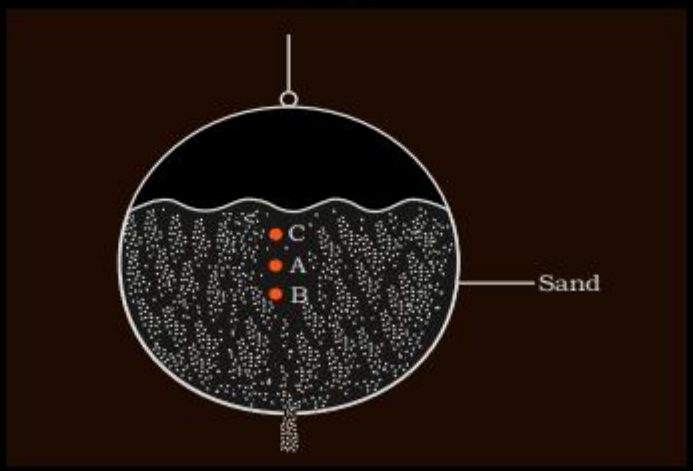


Fig. E 6.5: *Variation of centre of gravity of sand filled hollow bob on time period of the pendulum; sand being drained out of the bob in steps.*

ACTIVITY 8

AIM

To identify a diode, a LED, a transistor, an IC, a resistor and a capacitor from a mixed collection of such items.

APPARATUS AND MATERIAL REQUIRED

Multimeter, a collection of diode, LED, transistor, IC, resistor and capacitor.

PRINCIPLE

A diode is a two terminal device. It conducts when forward biased and does not conduct when reverse biased. It does not emit light while conducting.

A LED (light emitting diode) is also a two terminal device. It conducts when forward biased and does not conduct when reverse biased. It emits light while conducting.

A transistor is a three terminal device. The terminals represent emitter (E), base (B) and collector (C).

An IC (integrated circuit) is a multi-terminal device in the form of a chip. But some may have only three terminals, e.g. 7805, 7806, 7809, 7912.

A resistor is a two terminal device. It conducts equally in both directions.

A capacitor is a two terminal device. It does not conduct but stores some charge when dc voltage is applied.

PROCEDURE

1. Check the physical appearance of the component.
   (a) If it has four or more terminals and has the appearance of a chip (black rectangular block), then it is an IC.

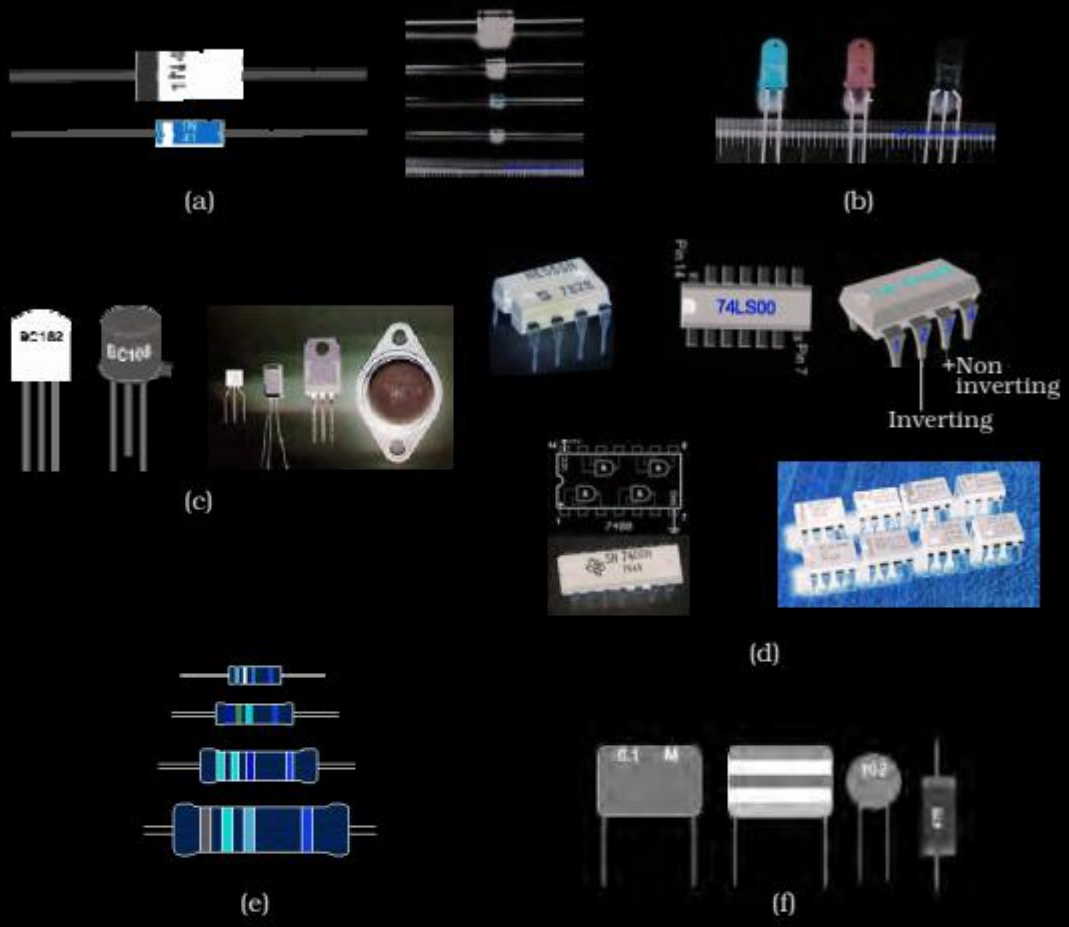


Fig. A 8.1 *(a) Diode (b) LED (c) Transistor (d) IC (e) Resistor (f) Capacitor*

(b) If it has three terminals, the component may be a transistor. To confirm, set-up the multimeter in resistance mode (highest range). Connect its black or common terminal to one of the extreme legs of the component and the second terminal (red or positive) to the central leg. Check the multimeter deflection. If a deflection is observed, interchange the multimeter terminals. If no deflection is observed, the component is a transistor. Repeat this test by connecting the multimeter terminals to the central leg and the other extreme leg. If a similar behaviour is observed, the component is a transistor.

(2) If the component has two terminals, it could be a resistor, a capacitor, a diode or a LED.

(a) Look for colour bands, if it has a typical set of three colour bands followed by a silver or gold band, the component is a resistor.

(b) Connect the multimeter terminals (in resistance mode-highest range) to the component terminals and watch for multimeter deflection. Also repeat by reversing the component terminals.

(c) If the multimeter shows an equal deflection in both the directions, the component is a resistor.

(d) If the deflection is accompanied with emission of light, in one direction and a much less or zero deflection in the other direction the component is a LED.

(e) If the multimeter does not show any deflection in one direction and shows deflection with no light emission in the other direction then, the component is a diode.

(f) If the multimeter does not show any deflection on connecting its terminals either way to the component, it is a capacitor. But if capacitance of capacitor is large, multimeter may show a momentary deflection.

(g) Record your observations in Table A 8.1 and A 8.2.

Note to teachers

(i) Avoid giving an IC with three terminals.

(ii) Preferably use analog multimeter and not the digital multimeter.

(iii) In case of digital multimeter, the term "deflection" in the instructions may be read as "reading".

(iv) Label each component by a different alphabet e.g., A, B, C, D, E...

OBSERVATIONS

Table A 8.1 : Checking for the number of terminals

Sl.No.	No. of terminals	Alphabet marked on the device	Name of the Device
1	Two		
2	Three		
3	More than three		

Table A 8.2 : Checking for the state of conduction

Sl.No.	State of conduction	Device Code	Name of the device
1	Conducts in one direction only without any emission of light		
2	Conducts in one direction only with emission of light		
3	Three terminal device; conducts in one direction only between central terminal and either of the remaining two terminals		
4	Conducts in both directions		
5	Does not conduct, gives an initial deflection which decays to zero		

RESULT

A diode, a LED, a transistor, an IC, a resistor and a capacitor are identified respectively from a mixed collection.

PRECAUTION

While obtaining resistance of any component, clean its leads properly.

SOURCES OF ERROR

1. When the metal ends of a multimeter leads are touched, the multimeter should show zero resistance. If it does not show it, bring the pointer to zero using 'Zero Adj knob' on the multimeter. If it is not done, the resistance measurements are not reliable.
2. While checking resistance of a component, avoid touching either of the metal ends of the multimeter leads. Body resistance in parallel with component resistance can affect the resistance measurement.

SUGGESTED ADDITIONAL EXPERIMENTS/ACTIVITIES

Studies of the basic circuit components such as resistor, capacitor may be undertaken on the following lines:

1. Resistors

Types of resistors

(a) *Wire wound resistors:* These are made by winding a specified length of wire of some conducting material. The material may be any alloy like mangnnin, constantan, nicrome etc.*

(b) *Carbon resistors:* They are made of a mixture of graphite (a form of carbon) and a good insulator like shellac in a specified proportion. The mixture is pressed and moulded onto rods by heating. By varying this proportion, carbon resistors are made for a very wide range of values. The stability of such resistors is poor but they are cheap and small.

(c) *Carbon film resistors***:* Carbon resistors are cheap and easily available. A thin film of carbon is deposited onto a small ceramic rod. The resistive coating is spiralled on it to get the desired value of resistance.

(d) *Thin film resistors:* These are made by depositing a very thin film of a conducting material on some insulating base. The thin film provides a very small cross-sectional area. Hence a large resistance. The films may be made from a metal or an alloy.

Tolerance of a resistor

The actual resistance value of a certain resistor may be different from its specified value. This may be due to external influences like changes in temperature, humidity etc. or due to the inherent limitation in manufacturing a given precise value. Deviations in the resistance value due to later cause is known as tolerance.

Wattage of a resistor

Every resistor has a maximum safe current that it can carry. Currents larger than that will produce excessive heat and may damage it. This is usually measured not in terms of the current, but in terms of maximum watt of power that can be dissipated in the resistor without causing any damage. It is known as its wattage. Common wattages for carbon resistors are 1/8, 1/2, 1 and 2 watts. Higher wattages are also available.

Colour code for resistor

Most common colour code used for resistor consists of three colour bands at one end.

* Wire wound resistors have more undesirable inductance than the other types of resistors.
**See appendix 3 for details.

The colour and their numerical meanings are:

Black – 0	Orange – 3	Blue – 6	White – 9
Brown – 1	Yellow – 4	Violet – 7	
Red – 2	Green – 5	Grey – 8	

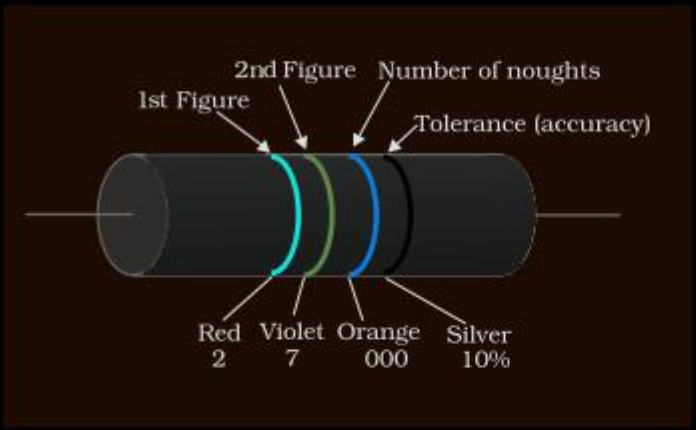


Fig. A 8.2 *A carbon resistor with colour code marking*

This colour code could be remembered easily by the term BBROY GB VGW (B B ROY Great Britain Very Good Wife). To read the resistance of a colour coded resistor, start with the strip nearest to the end. The colour of the first strip gives the first digit in the resistance value. The colour of the second strip stands for the second digit. The colour of third strip indicates the value of multiplier or the number of zero following second number.

Resistors with only three bands have a tolerance of 20%. Its actual value may vary 20% up or down from its indicated value. If a fourth band is added separated from the first three, then the tolerance of the resistor is known from the colour of the fourth band. If the colour of the fourth band is silver, then tolerance is 10%. Gold colour represents 5% tolerance red represents 2% and brown represents 1%. The ohmic values are often printed on 2% and 1% resistors.

2. Capacitor

Capacitor is a device that stores electrical charge. A capacitor blocks the passage of dc while it allows ac to flow through it.

There are many types of capacitors having different types of dielectrics in general use. These are

(i) Air capacitor (variable gang capacitors)

(ii) Mica capacitor (low capacitance)

(iii) Ceramic capacitor (very low capacitance)

(iv) Paper capacitor (low capacitance)

(v) Plastic capacitor

(vi) Electrolytic capacitor (medium capacitance)

(vii) Oil filled capacitor (high capacitance)

3. Diode

In some semiconductor junction diodes, a silver ring is shown on one of its end. This end of the diode is n-side and the other end is p-side [Fig. A 8.3(a)]. In some diodes, the symbol of the diode is painted on the body of the diode. The direction of arrow is the direction of current flow. Hence, the side from which the arrow starts is p-side and the side towards which the arrow ends is n-side [Fig. A 8.3 (b)]. Some diodes are bullet shaped in which case the flat side is p-side and the cylindrical side is n-side [Fig. A 8.3 (c)]. In case of a light emitting diode (LED), usually the shorter pin is n-side and the longer pin is p-side [Fig. A 8.3 (d)].

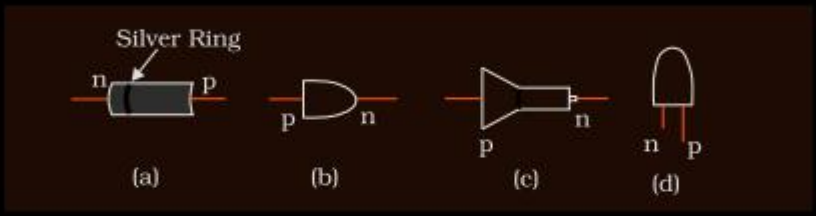


Fig. A 8.3 *Shapes of some diodes*

4. Transistors

In this activity, a method to distinguish between n-p-n and p-n-p type transistors has been given, provided the pin diagram of the transistor is known. In case, if no pin diagram is known, one can still determine the type (n-p-n or p-n-p) of the given transistor. This can be done by measuring the resistance values between different terminals of the given transistor [(Fig. A 8.4) (a) and (b)]. Table A 9.5 (p.181) summarises the nature of resistance values for n-p-n and p-n-p transistors.

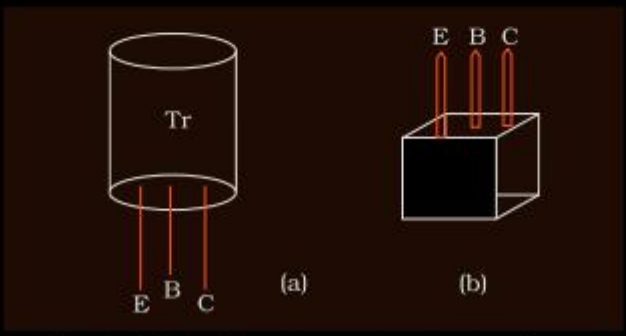


Fig. A 8.4 *Different transistor terminals*

ACTIVITY 9

AIM

Use of Multimeter to

(A) check whether the diode is in working order and to check unidirectional flow of current in a diode; (B) identify emitter, base collector of a transistor; (C) distinguish between n-p-n and p-n-p transistors and check whether the transistor is in working order.

APPARATUS AND MATERIAL REQUIRED

A multimeter, a diode, a transistor (whose base diagram is known), plug key, a battery, resistance box, connecting wires and sand paper.

PRINCIPLE

Analog multimeter used in resistance mode

In this selection a circuit with a battery and resistors come into play. The circuit diagram below (Fig. A 9.1) indicates the main circuit components: a 3V battery (provided in the multimeter), a fixed resistance, a rheostat and a galvanometer G. Red and black leads are connected at fixed resistance ends B and A respectively.

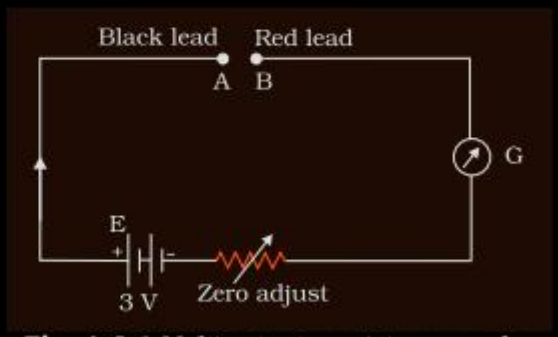


Fig. A 9.1 *Multimeter in resistance mode*

The deflection is adjusted to full scale deflection (zero mark on the scale), using the rheostat (zero adjust) by first shorting the red and black leads. Any resistance which has to be tested/evaluated is placed between A and B. The deflection due to current in the circuit is calibrated to read the resistance.

You will notice that the black lead is actually connected to the positive and red to the negative of the internal 3 V battery. While measuring values of resistances, it is immaterial how they are connected across A and B. However, it becomes important when this battery also provides the bias for a diode or a transistor junction.

(A) Checking whether a diode is in working order and whether flow of electric current is unidirectional through it

A semiconductor junction diode, when connected (or biased) in the forward direction, offers a low resistance. However, when it is biased in the reverse direction, it offers a very high resistance. Therefore, the working of a junction diode can be examined by measuring its resistance in the forward and reverse bias conditions. The functioning of junction diode can also be examined by checking the unidirectional flow of current in a circuit as shown in Fig. A 9.2. For a diode in proper working order, a substantial current (of the order of few mA) will flow in the forward bias condition. If the polarity of the bias is reversed (i.e., diode in reverse bias condition), negligible current (of the order of few μA) will flow.

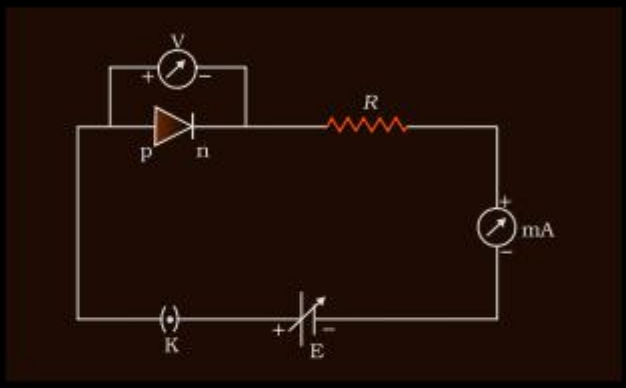


Fig. A 9.2 *A semiconductor junction diode in the forward bias*

(B) Identification of emitter, base and collector of a transistor

In general, each transistor has three pins of the same size. In some transistors a dot or mark is put on the body near one of the pins. This pin is the emitter. The Emitter (E), Base (B) and Collector (C) of a transistor can be identified using its pin diagram. Fig. A 9.3 shows the base diagram of a typical metal capped cylindrical transistor. This has been drawn by keeping the pins of transistor pointing upwards. In fact, there is no universal pin diagram for the transistors. One can find the pin diagram of a given transistor (transistors have code numbers AC 127, BC548, 2N3055HL 100 etc.) in most transistor manuals.

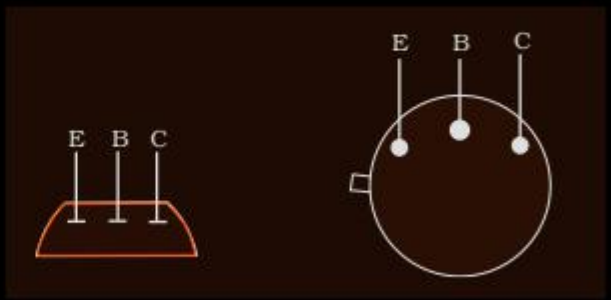


Fig. A 9.3 *Base diagram of a typical metal-capped cylindrical transistor. Note that the pin connections may be different for different transistors. It is advisable to refer to the manufacturer's data sheet.*

(C) Distinguishing between n-p-n and p-n-p transistors and checking whether the given transistor (p-n-p or n-p-n) is in working order

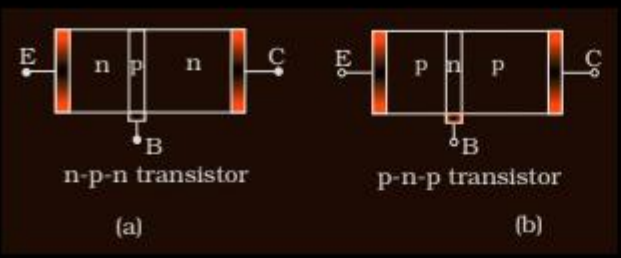


Fig. A 9.4 *Schematic representation of (a) n-p-n, and (b) p-n-p transistors*

The base B of an n-p-n transistor is p-type material whereas emitter E and collector C are of n-type materials (Fig. A 9.4 a). Therefore, the forward resistance between the base B and the emitter E (or collector C) should have a low value. In case of a p-n-p transistor (Fig. A 9.4 (b) (whose emitter E, base B, and collector C are p-, n- and p-type materials respectively), the forward resistance between the emitter and base should be low.

The resistances between the base-emitter, and base-collector are measured. For a transistor in working order, resistance values follow the results as shown in Table A 9.1.



Table A 9.1 : Resistance values between different terminals of n-p-n and p-n-p transistors

Sl.No.	One of the transistor terminals (B, C or E) connected to negative lead (black) of the multimeter	Transistor terminal connected with positive (red) lead of the multimeter	Bias	Resistance
A. For n-p-n transistor				
1	E	B	Forward	Low
2	C	B	Forward	Low
3	B	C	Reverse	Very high
4	B	E	Reverse	Very high
B. For p-n-p transistor				
1	B	E	Forward	Low
2	B	C	Forward	Low
3	E	B	Reverse	Very high
4	C	B	Reverse	Very high

Any deviation from the results shown in Table A 9.1 would indicate that the given transistor is not in working order. For example, for an p-n-p transistor, a low resistance value between terminal E (when connected with the negative lead of the multimeter) and terminal B (when connected with positive lead of the multimeter) indicates that the transistor's terminals E and B are short-circuited. For a working transistor, it should have shown very high resistance, being reverse biased.

PROCEDURE

1. The diodes, transistors, and connecting wires may have some insulating layers on their surface if they are not used for a long time. Therefore, clean their ends using a sand paper till they shine.

(A) Checking whether a diode is in working order and whether flow of electric current is unidirectional through it

2. Set the multimeter in resistance measuring mode.

3. Connect the diode terminal 1 with the positive lead of the multimeter and terminal 2 with the negative lead of the multimeter. Measure the diode resistance. Reverse the diode connections and again measure the diode resistance. Record your observations in Table A 9.2. Also conclude whether the given diode is in working order or not.

Note: A low value of diode resistance (ranging from few Ω to kΩ) indicates that the diode is connected in forward bias. While a very high value of resistance (of the order of MΩ), indicates that the diode is connected in reverse bias.

If both the resistance values, i.e., in reverse and forward directions, are low then it means that the diode is short-circuited. On the other hand if both the resistance values are very high, then the diode junction may be considered as discontinuous or open. Thus, in both the conditions, the diode condition may be concluded as not in working order.

4. Identify the p-and n-terminals of the given diode from the observations.

5. Connect the diode with the cell and resistance box in a series circuit (without key ON) as shown in Fig. A 9.2. Set the multimeter in current measuring mode at a suitable range (mA, start from high current range).

6. Take out a suitable resistance R from the resistance box in the circuit to restrict the current within the current range chosen (in the multimeter). Put the key ON and measure the current flowing through the circuit. Record your observations in Table A 9.3.

7. Repeat step 6 for a few more values of resistance R in the circuit.
8. Reverse the polarity of diode (now the diode is reverse biased) and repeat steps 6 and 7.

(B) and (C) Identifying n-p-n and p-n-p transistors and checking whether the given transistor (p-n-p or n-p-n) is in working order

9. Identify the emitter E, base B, and collector C of the given transistor by observing dot or a mark on the body of transistor near one of its pins. Draw the base and schematic diagrams of the given transistor in your notebook.
10. Set the multimeter in 'resistance measuring mode'.
11. Connect the positive (red) lead of the multimeter with the base and negative led (black) with emitter (or collector) terminal and measure the resistance. Record the observations in Table A 9.4.
12. Reverse the connections of the multimeter and again measure the resistance values between the transistor's base B and emitter E (or collector C) terminals. Record observations in Table A 9.4.
13. From Table A 9.4, check whether the given transistor is in working order or not.
14. For a transistor that is in working order, using the information given in Table A 9.1, infer the nature of the given transistor (n-p-n or p-n-p).

OBSERVATIONS

1. Table A 9.2 : Measurement of diode resistance

Sl. No.	Diode terminal connected with the positive lead of the multimeter	Diode terminal connected with the negative lead of the multimeter	Resistance (Ω)
1	1	2	
2	2	1	

2. Table A 9.3 : Unidirectional flow of current through a diode

(a) When diode is connected in forward bias

Sl. No.	Resistance (Ω)	Current (mA)
1		
2		

(b) When diode is connected in reverse bias

Sl. No.	Resistance (Ω)	Current (mA)
1		
2		
3		

3. Table A 9.4: Resistance values between different transistor terminals

Sl. No.	Transistor terminal connected with the positive lead of the multimeter	Transistor terminal connected with the negative lead of the multimeter	Resistance (Ω)
1	B	E	
2	B	C	
3	E	B	
4	C	B	

RESULT

1. The p- and n-sides of the given diode are identified from Table A 9.2.
2. The given diode is/is not in working order (from Tables A 9.2 and A 9.3)
3. The unidirectional flow of current in a diode has/has not been realised (from Table A 9.3).
4. The three terminals of the given transistor have been identified and the pin diagram has been shown in Fig. A 9.3.
5. The given transistor is n-p-n/p-n-p (from Table A 9.4).
6. The given transistor is/is not in working order (from Table A 9.4).

PRECAUTIONS

1. While obtaining resistance of any component, clean its leads properly.
2. Use the selector switch of the multimeter carefully for various measuring modes.

SOURCES OF ERROR

1. When the metal ends of multimeter leads are touched, the multimeter should show zero resistance. If it does not show this, bring the pointer to zero using 'zero Adj knob' on the multimeter. If it is not done, the resistance measurements are not reliable.
2. While checking resistance of a component, avoid touching both the metal ends of the multimeter leads. Body resistance in parallel with component resistance can affect the resistance measurement.

DISCUSSION

1. In a transistor, the junction joining the base region and the emitter region is called base-emitter junction. The junction joining the base region and the collector region is called base-collector region.
2. When multimeter is in current measuring mode, shunt resistance of different values are introduced in the circuit paralleled with the coil. In voltage measuring mode resistances of different values come in contact with the circuit in series with the coil. When you select the switch of the multimeter for measurement of resistance, different resistances are introduced in circuit in series with the cell of the multimeter.

SELF ASSESSMENT

Find the base of a transistor if pin diagram is not available. You may use the Table A 9.5 for this purpose.

Using the results given in Table A 9.5, it can be determined whether the given transistor is n-p-n or p-n-p type.

Table A 9.5 : Nature of resistance values between different transistor terminals (when pin diagram is not known)

(Note: Here terminal 2 is assumed as the base pin of transistor)

Sl. No.	Transistor terminal connected with the positive lead of the multimeter	Transistor terminal connected with the negative lead of the multimeter	Nature of resistance
A. For n-p-n transistor			
1	1	2	Very high
2	1	3	Very high
3	2	1	Low
4	2	3	Low
5	3	1	Very high
6	3	2	Very high
B. For p-n-p transistor			
1	1	2	Low
2	2	1	Very high
3	1	3	Very high
4	3	1	Very high
5	2	3	Very high
6	3	2	Low